

Biology Detailed Practice Workbook

Concepts • Questions • Mini Mock

Part 1 — Cell Biology

- Name the organelle responsible for ATP production in eukaryotic cells.
- State one major difference between prokaryotic and eukaryotic cells.
- Arrange the major phases of the cell cycle in order.
- Differentiate mitosis and meiosis using chromosome number and biological purpose.
- Identify the cell structure responsible for selective permeability.

Part 2 — Genetics & Molecular Biology

- Find the expected phenotypic ratio of a simple monohybrid cross under complete dominance.
- Explain the meaning of genotype and phenotype.
- Write the complementary DNA bases for a short DNA sequence.
- Differentiate transcription and translation.
- Explain why mutations can contribute to genetic variation.

Part 3 — Plant & Human Physiology

- List the major stages involved in photosynthesis at a conceptual level.
- Explain the direction of water movement through a plant.
- Trace the path of blood through pulmonary circulation.
- State the functional role of the nephron.
- Match a major endocrine gland with one hormone it produces.

Part 4 — Ecology & Biotechnology

- Construct a simple food chain and identify each trophic level.
- Explain why energy decreases across trophic levels.
- Differentiate biodiversity conservation in natural and managed settings.
- State the purpose of a restriction enzyme in recombinant DNA work.
- Give examples of biotechnology applications in medicine or agriculture.

Mini Mock

1. The basic structural and functional unit of life is: A) tissue B) cell C) organ D) organism.
2. DNA base pairing includes: A) A-G B) A-T C) C-T D) G-T.
3. The structural and functional unit of kidney is: A) neuron B) nephron C) alveolus D) villus.

- 4. The 3:1 ratio is classically associated with: A) monohybrid phenotypic inheritance B) translation C) respiration D) ecology.
- 5. Energy in an ecosystem generally: A) increases at every trophic level B) remains identical C) decreases between trophic levels D) is created by consumers.
- Answers: 1-B, 2-B, 3-B, 4-A, 5-C.